

## Architectural Specification

# GLAZING

### SECTION 8.10

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## SECTION 8.10

### GLAZING

#### PART 1 GENERAL

##### 1.01 DESCRIPTION

- A. The Work of this Section includes, but is not limited to, the furnishing and installing of the following:
1. Glass, decorative glass, mirror products and materials as indicated on the Contract Documents.
  2. Glazing compounds & high-performance sealants.
  3. Glazing accessories & fixings.
  4. Fire Rated Glass

##### 1.02 REFERENCES

A. American Standards

Except as modified by governing codes and by the Contract Documents, comply with the applicable provisions and recommendations of the following:

1. GANA "Sealant Manual" and "Glazing Manual".
2. GTA "Engineering Standards Manual".
3. NAMM "National Association of Mirror Manufacturers "Mirrors, Handle with Extreme Care, Tips for the Professional on the Care and Handling of Mirrors."
4. 4. GANA "Laminated Glass Design Guide."
5. ANSI Z97.1 - Safety Performance Specifications and Methods of Test for Safety Glazing Materials Used in Buildings.
6. ASTM C920 : 1987 - Specification for Elastomeric Joint Sealants.
7. ASTM C1193 : 1992 - Guide for Use of Joint Sealants.
8. ASTM C1036 - Specification for Flat Glass.
9. ASTM C1048 - Specification for Heat-Treated Flat Glass.
10. ASTM E7733 - Test Method for Seal Durability of Sealed Insulating Glass Units.
11. CPSC 16 CFR 1201 - U.S. Consumer Product Safety Commission Standard on Architectural Glazing Materials.
12. Flat Glass Marketing Association (FGMA) - GM-Glazing Manual and Glazing Sealing Systems Manual.
13. Safety Glazing Standard: Where safety glass is indicated or required by authorities having jurisdiction, provide type of products indicated which comply with ANSI Z97.1 and testing requirements of CPSC 16 CFR Part 1201 for category II materials. Subject to compliance with requirements and local authorities having jurisdiction, provide safety glass with a removable certification label of Safety Glazing Certification Council (SGCC) or other certification agency acceptable to authorities having jurisdiction.

B. British Standards

1. BS 952 - Glass for Glazing.
2. BS5516: 1991 - Code of Practice for Design and Installation of Sloping and Vertical Patent Glazing.
3. BS5713: 1979 - Specification for Hermetically Sealed Flat Double Glazing Units.
4. BS5889: 1989 - Specification for One-Part Gun Grade Silicone-Based Sealants.
5. BS6206: 1981 - Specification for Impact Performance Requirements for Flat. Safety Glass and Safety Plastics for Use In Buildings.
6. BS6262: 1982 - Code of Practice for Glazing for Buildings.
7. BS8000 : Part 7 : 1990 - Code of Practice for Glazing.

C. Hong Kong Standards

1. Cap. 123 Building (Construction) Regulation.

- D. Where the language in any of the documents referred to herein is in the form of a recommendation or suggestion, such recommendations or suggestions shall be deemed to be mandatory under this Contract.

**1.03 QUALITY ASSURANCE**

- A. Glazier Qualifications: Engage an experienced glazier who has completed glazing similar in material, design, and extent to that indicated for this Project and with a record of successful in-service performance.
- B. Sample of each type of glass should be submitted and compatible to control sample or equivalent approved by the Architect.
- C. Notify the glass manufacturer of the mock up test program and provide the opportunity to the manufacturer to observe the tests and instrument the mock up(s), should he so desire, provided such participation causes no delays and is solely at the expense of the glass manufacturer.
- D. Smart Glass:
1. The work shall be performed by a firm having 5 years experience in the installation of materials specified herein on projects comparable to this Project. The firm shall have the approval of the materials manufacturer.
  2. Single-source Responsibility: Obtain the smart glass from a single source with resources to provide products of consistent quality in appearance and physical properties without delaying progress of the Work.
  3. Qualification of Manufacturers: Company specializing in the manufacturer of smart glass with minimum of ten years documented experience.

**1.04 PERFORMANCE REQUIREMENTS**

- A. Unless otherwise specified, all materials shall be of the best quality of the types specified.
- B. All glass for glazing shall generally conform to B.S. 952: Part 1: 1995 and B.S. 952: Part 2:1980.
- C. All glass shall be of accurate size with clean undamaged edges and surfaces which are not disfigured. All glass shall be free from the following :
1. Bubbles,
  2. Smoke wanes,
  3. Air holes,
  4. Scratches,
  5. Other defects.

- D. All glass shall be cut to fit the rebates with due allowance made for expansion.
- E. Glass panes shall be kept dry and clean during delivery and shall be stored vertically in a well vented location carefully protected from condensation and other moisture. There must be air between stored glass sheets.
- F. The thickness of glass shall meet with the design wind pressure or suction as specified hereinafter or indicated on drawings.
- G. Any glass not conforming to the requirements of the Specification shall be rejected and removed from site.
- H. All glass shall be cut to fit the rebates with due allowance made for expansion.
- I. The dimensions of the glazing shown on the drawings shall be used as a reference. The contractor shall verify on site prior to preparation of shop drawings.
- J. Workmanship
  - 1. General Requirements
    - a) Glazing shall conform to B.S. 6262: 1982 and B.S. 8000: Part 7: 1990 generally.
    - b) Unless otherwise indicated on drawings, each opening shall be glazed with one pane of glass.
    - c) All directionally patterned or wired glass shall be fixed with the pattern or wires parallel to the surround.
    - d) Align patterns or wires in each adjacent pane.
    - e) Cut edges of wired glass shall be painted with black bituminous paint to inhibit rusting.
    - f) Proper allowance shall be made for thermal expansion during cutting of glass.
    - g) Form openings in glass to exact sized as specified and with edges ground smooth to remove sharp arises.
    - h) External glazing shall be wind tight and watertight on completion. Cast or figure rolled glass for external glazing shall be fixed with patterned side inside.
    - i) Translucent rolled glass for external glazing shall be fixed with patterned side inside.
    - j) Store glass in crates until just prior to installation. Replace any glass with breaks or sustains edge or surface damage.
      - i. Flare shall not occur at setting blocks. Flare elsewhere shall not exceed 1mm, measured across the edge of the glass.
      - ii. Shark teeth shall not penetrate deeper than half the glass thickness.
      - iii. Serration hackle shall not occur within 150mm of corners.
      - iv. Flake chips are chips with a depth not greater than 0.8mm or length or diameter greater than 6mm chips that exceed these dimensional limits are classified as rough chips. Rough chips are not permitted.
      - v. Bevel shall not be greater than 1.6mm.
  - 2. Preparation of Surrounds
    - a) Rebates shall be clean, dry and unobstructed when setting the glass. Thoroughly clean glazing pocket before setting glass. Ensure that

solvents used are compatible with finished aluminium, glass and other glazing materials.

- b) Timber rebates shall be suitably primed before glazing with linseed oil putty.
  - c) Metal surrounds shall be prepared to the recommendations of the manufacturer of the glazing materials to be used and to the recommendations of the component manufacturer.
3. Bedding
- a) Glazing shall be carried out to ensure that no voids or spaces are left in bedding compound and that there is metal or wood to glass contact.
  - b) Surplus bedding compound shall be stripped as follows:-
    - i. To top and side edges - flush with top of rebate of bead.
    - ii. To bottom edges - at 45o angle.
  - c) Open lights shall be left in close position until the bedding compound has set with sufficient strength to prevent displacement of glass.
4. Setting and Location Blocks
- a) Setting and location blocks shall be used for all glass panes exceeding 0.2 square meter and located in accordance with B.S. 6262 or as recommended by the window manufacturer, but not more than 75mm and less than 30mm from corner of surround to end of block nearest corner. Setting and location blocks shall be used for all solar control glass.
5. Glazing with Putty
- a) Secure glass to wood surrounds with glazier's sprigs at 450mm (maximum) centres.
  - b) Secure glass to steel surrounds with spring clips as provided by or recommended by the surround manufacturer.
  - c) Black putty shall be of regular thickness and 2mm (minimum).
  - d) Form front putty to neat triangular fillet, stopping 2mm short of sight line.
  - e) Leave opening lights in closed position until putty has set sufficiently to prevent displacement of glass.
6. Protection of Putty
- Seal linseed oil putty and conventional metal window putty as soon as sufficiently hard and within the time recommended by their manufacturers, with primer and paint as specified in "Painting".
7. Glazing with Beads
- a) For external glazing, bed external beads in glazing compound.
  - b) Secure wood beads with rustproofed panel pins or with countersunk brass screws and cups at 200mm (maximum) centres and 75mm (maximum) from each corner.
  - c) Remove and refix metal beads supplied with metal windows.
  - d) When required, glaze with plastics glazing channel, glazing compound and beads.
8. Sealing
- a) When specified, seal the joint between the glass and bead with an approved silicone sealant. Do not apply sealants when substitutes are wet or the temperature is below 4oC.
9. Preformed Taped and Gaskets

- a) Obtain approval to use preformed tapes, shims and compression gaskets for glazing in accordance with the manufacturer's recommendations.
10. Edge Clearances
- a) Edge clearances shall be equal all round each pane, and 3mm (minimum).
  - b) For solar control glass, the glazing clearance shall be a minimum of 3mm all round for panes up to 750mm (30 in.) major dimension and a minimum of 5mm all round for panes 750mm and over in major dimension.
11. Edge Cover
- a) Edge cover provided by glazing compound shall be as follows :-

| Glass area (m2) | Edge cover (mm)                                  |
|-----------------|--------------------------------------------------|
| Up to 0.5       | 5                                                |
| 0.5 to 1.5      | 9                                                |
| 1.5 to 4.0      | 12                                               |
| Over 4.0        | As recommended by glazing compound manufacturer. |

- b) As recommended by glazing compound manufacturer.
12. Glazing Aluminium Windows
- a) Aluminium windows shall be glazed by the specialist supplying and fixing the windows.
13. Mirrors
- Unless otherwise shown on the drawings mirrors shall be fixed on 12mm marine plywood board with waterproof building paper backing. All edges shall be sealed with Shellac prior to fitting. Where so required mirrors shall be drilled for and fixed with fixing screws and chromium plated dome headed screws and neoprene grommets.
14. Glazing Plastic Sheets
- a) Cut plastic sheets with a fine tooth saw and smooth or chamfer edges. Allow edge clearance and edge cover as recommended by the manufacturer. Only remove sufficient of the protective covering around edges to allow glazing and remove the remainder just prior to substantial completion.
15. Sealing Glass to Glass Joints
- a) Sealant shall be applied in accordance with the manufacturer's instructions. Glass surfaces or either side of the joints to be sealed shall be properly taped and protected, and all superfluous sealant cleared off to leave an even, parallel sided joint.

16. Glass Louvres

- a) Glass blades to fixed and adjustable louvres shall be 6mm thick with exact widths and lengths as shown on the drawings.
- b) Glass blades shall have edges truly parallel and ground to remove sharp arises.
- c) In adjustable louver window frames, glass blades shall be fixed into clips as recommended by the frame manufacturer.

17. Fire Resistant Glass

- a) Fire resistant glass used in fire doors, fixed windows and partitions shall be so constructed and of the thickness as recommended by the manufacturer and in accordance with the requirements of fire rating as specified.
- b) Fire resisting performance requirement for all F.R.R. windows shall be not less than 120 minutes in terms of integrity and insulation in accordance with BS EN 1364-1:2015.
- c) The Sub-Contractor shall submit shop drawing showing the glass thickness, construction details and glazing materials to the Architect for his approval and shall also submit test certificates from the manufacturer and letters of approval from the Government recognizing that the said construction and materials comply with the requirement of B.S. 476 : Part 22 : 1987.
- d) Impact safety rating must be at least B according to BS 6206.
- e) Timber for glazing beads must be hardwood with a minimum density of 640kg/m<sup>3</sup>, unless the proprietary system can demonstrate adequate performance with alternative materials using the appropriate glass types and glazed areas. Beads must be of the appropriate size and profile as dictated by the manufacturer.
- f) Glazed apertures must not be nearer than 100mm to any leaf edge. Multiple apertures are acceptable up to the maximum approved area with a minimum dimension of 80mm core separating the apertures.
- g) Fire resistant glass products must demonstrate adequate performance in the required pane size when tested in doorsets of comparable construction. The Contractor shall provide documentary evidence to satisfy the requirements. Glass proven only by tests in screens are not acceptable; false beads must not be applied across the glass face.

18. Cleaning and Making Good

- a) Remove all smears and excess glazing compound.
- b) Remove mortar, plaster or concrete spillage and paint drips whilst wet.
- c) All glass and glazing materials shall be thoroughly washed on both sides and left clean on completion. For special glass (e.g. reflective glass), cleaning shall be performed in strict accordance with glass manufacturer's instruction.
- d) All glass shall be free from scratches, cracks, etc.
- e) Replace glasses or fixing materials which have been broken or damaged and re-decorate the attached surrounding finishes before completion of the works.

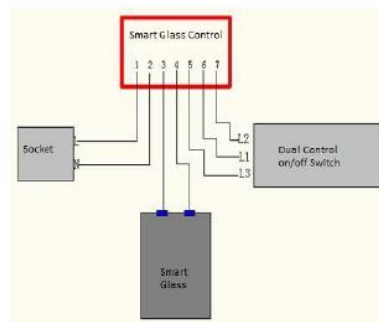
## K. Smart Glass:

### Specification of STG

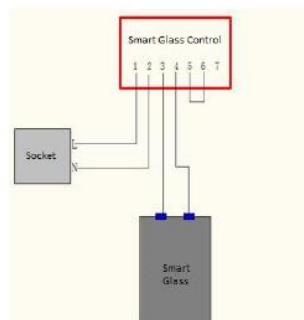
| 项目 Item                                 | 模式 Mode                               | 参数 Specification                |
|-----------------------------------------|---------------------------------------|---------------------------------|
| 光学参数<br>Optical<br>Characteristic       | 可见光透射比<br>Visible Light Transmission  | On >80%                         |
|                                         |                                       | Off >65%                        |
|                                         | 平行光透射比<br>Parallel Light Transmission | On >78%                         |
|                                         |                                       | Off <1%                         |
|                                         | 雾度 Haze                               | On 5%                           |
|                                         |                                       | Off 95%                         |
| 电学参数<br>Electrical<br>Characteristic    | 可视视角<br>Viewing Angle of Screen       | On 140°                         |
|                                         | 紫外阻隔率<br>UV Blocking                  | On&Off ≥ 98%                    |
|                                         | 工作电压<br>Operating Voltage             | On 48V/65V AC                   |
|                                         | 响应时间<br>Response Time                 | Off=>On <10ms<br>On=>Off <200ms |
| 机械参数<br>Mechanical<br>Characteristic    | 功耗<br>Power Consumption               | On <4 w/ m²                     |
|                                         | 厚度<br>Thickness of STG                | 10,12,14,18mm                   |
| 环境参数<br>Environmental<br>Characteristic | 单片最大尺寸<br>Max size                    | 4200*2100mm                     |
|                                         | 工作温度<br>Working Temperature           | -10~ 60℃                        |
|                                         | 储存温度<br>Storing Temperature           | -20~ 60℃                        |
| 隔音 Sound Insulation                     |                                       | 35dB                            |
| 使用寿命 Life Time                          |                                       | On >80000 小时                    |
| 开关次数 Switch Frequency                   |                                       | >200 万次                         |
| 质保期 Guarantee                           |                                       | 12 个月                           |

### Control Options

#### Dual Control



#### Single Control





## **1.05 ENVIRONMENTAL REQUIREMENTS**

(NOT USED)

## **1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS**

Do not perform glazing when temperature is below 40 deg. F., unless the manufacturer of the glazing materials specifically recommends application of his materials at lower temperatures. If job progress or other conditions require glazing work when temperatures are below 40 deg. F. (or below the minimum temperature recommended by the manufacturer), consult the manufacturer and establish the minimum provisions required to ensure satisfactory work. Record in writing to the manufacturer, with copy to the Architect, the conditions under which such glazing work was performed and the provisions made to ensure satisfactory work.

## **1.07 SUBMITTALS**

- A. Submittals as indicated in Section 1.5.
- B. Samples: Submit 3 samples of each type of glass, setting block, face shim, edge block, glazing sealant and gasket. Provide sealant samples 300mm long installed between samples of the materials to be glazed, fully cured. Samples will be reviewed for color and texture only.
- C. Glass samples to be 300x300mm in size and clearly identified.
- D. Glazing Schedule: Provide a glazing schedule utilizing the same designations shown on Drawings for glazed openings listing glass types and thicknesses for each size opening and location.
- E. Product Data: Submit the following information from the manufacturer:
  - 1. Statement that each product to be furnished is recommended for the application shown.
  - 2. Product Test Reports: Provide product test reports from a qualified testing agency indicating the following products comply with requirements, based on comprehensive testing of current products:
    - a) Glazing sealants.
  - 3. Complete instructions for handling, storing, mixing, priming, installing, curing and protecting each glazing material.
  - 4. Mirror Product Data
    - a) Submit product data for mirrors including description of materials and process used to produce mirrored glass, including source of glass, glass coating components, edge sealer, and quality control provisions.
    - b) Mirror mastic glass coating compatibility test reports from organic protective coating manufacturer indicating that mirror mastic has been tested for compatibility and adhesion with organic protective coating. Include organic coating manufacturers' interpretation of test results relative to performance and recommendations for use of mastics with organic protective coating.
- F. Statement of Application: Submit copy of statement signed by the Subcontractor and Contractor, stating that the glazing materials complied with these Specifications and that the installation methods complied with the manufacturer's printed instructions for each condition of use on this Project.
- G. Manufacturer's calculations of the glass thickness.

- H. Submit Statement of Manufacturers Review Form.
- I. Product, Physical and technical data for each type of glass.
- J. Maintenance Instructions: Furnish maintenance instructions for each type of glass for use during construction and for use by the Employer after acceptance of the Work.
- K. Any fabrication of glazing conducted outside of Hong Kong Special Administrative Region shall be required to comply with the standards within this specification including but not limited to all British Standards and Code of Practice, circulars, guidelines and specifications issued by the relevant government departments. For fabrication conducted outside of Hong Kong Special Administrative Region, the contractor shall bear:
  - 1. all related costs (including transportation, accommodation and insurance) required for the Architect's appointed representative to carry out and complete the inspection; and
  - 2. any cost related to the appointment of an independent checker from a HOKLAS accredited laboratory or equivalent to be approved by the Architect, to assure materials have been checked, calibrated and certified within acceptable parameters according to analytical, safety and quality control procedures stipulated by all relevant specified government and industry standards as agreed with the Architect.
- L. Prepare sample installation before proceeding with the work. Glaze sample in the presence of the glazing materials manufacturer and glass manufacturer, not less than one (1) week in advance of the scheduled glazing work.

#### **1.08 PRE-INSTALLATION CONFERENCE**

(NOT USED)

#### **1.09 DELIVERY STORAGE AND HANDLING**

- A. Deliver glazing materials to Project site in manufacturers' unopened containers, fully identified with trade name, color, size, hardness, type, class, and grade. Store each item in accordance with manufacturer's instructions.
- B. Deliver, store and handle glass in accordance with manufacturer's recommendations; protected from weather, staining and damage. During storage and handling of glass provide cushions at edges to prevent impact damage. Protect glass from scratches and abrasion

#### **1.10 MOCK-UPS**

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

#### **1.11 TESTING**

(NOT USED)

#### **1.12 WARRANTY**

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

- B. Warranties specified in this Article shall not deprive the Employer of other rights the Employer may have under other provisions of the Contract Documents and are in addition to and run concurrent with other warranties made by the Contractor under requirements of the Contract Documents.
- C. Laminated and Tempered Glass: Provide a ten (10) years written warranty, directly to the Employer, warranting against deterioration of laminated and tempered glass. Deterioration of laminated and tempered glass is defined as the development of manufacturing defects including edge separation or delamination that materially obstructs vision through glass and blemishes exceeding those allowed by referenced laminated-glass standard. Upon notification of such defect, within the warranty period, make the necessary replacements at the convenience of the Employer.
- D. Mirror: Provide a ten (10) years written warranty, directly to the Employer, warranting against any evidence of silver spoilage. Upon notification of such defects, within the warranty period, make the necessary replacements at the convenience of the Employer.
- E. Silicon Sealants: Provide a ten (10) years written warranty directly to the Employer for all high-performance silicon sealants.
- F. FRR Glass, Provide a ten (10) years written warranty directly to the Employer warranting against manufacturing defect and change in properties. Upon notification of such defect, within the warranty period, make the necessary replacements at the convenience of the Employer.

#### **1.13 SPARE PARTS**

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

## PART 2 PRODUCTS

### 2.01 APPROVED MANUFACTURER/SUPPLIERS

(NOT USED)

### 2.02 MATERIAL

#### G. Clear Sheet Glass

1. Clear sheet glass shall be transparent glass drawn of "Ordinary Quality" (O.Q.) for glazing.
2. Clear sheet glass shall be of the following nominal thickness and approximate weight :-

| <u>NOMINAL THICKNESS</u> | <u>APPROX. WEIGHT</u> |
|--------------------------|-----------------------|
| 3 mm                     | 7.5 kg/sq.m.          |
| 4 mm                     | 10.0 kg/sq.m.         |
| 5 mm                     | 12.5 kg/sq.m.         |
| 6 mm                     | 15.0 kg/sq.m.         |

#### H. Obscured (Translucent) Glass

1. Obscured glass shall be translucent glass cast or figure rolled of the type and quality for glazing.
2. Obscured glass shall have the required degree of obscuration and diffusion.
3. Obscured glass shall generally be used for windows to all toilets and bath rooms.
4. Obscured glass shall be of the following nominal thickness and minimum weights:-

| <u>NORMAL THICKNESS</u> | <u>MIN. WEIGHT</u> |
|-------------------------|--------------------|
| 3 mm                    | 6.0kg/sq.m.        |
| 4 mm                    | 7.5 kg/sq.m.       |
| 5 mm                    | 9.5 kg/sq.m.       |
| 6 mm                    | 11.0 kg/sq.m.      |
| 10 mm                   | 21.5kg/sq.m.       |

#### I. Float or Polished Plate Glass

1. Float glass unless otherwise specified shall be clear float annealed glass. Float glass shall be manufactured in continuous ribbon from floated in molten condition upon liquid metal at controlled temperatures. The glass shall be completely transparent glass with both surfaces flat and parallel to provide clear, undistorted vision and reflection.
2. Polish plate glass for glazing purposes shall be of "Glazing Glass" (G.G.) quality with both surfaces ground, smoothed and polished to provide clear, undistorted vision.
3. Generally, clear float glass shall be used in lieu of polished plate glass in thickness up to 25mm. Polished plate glass is still available in thicknesses greater than 25mm and up to 38mm.
4. Clear float glass shall be of the following nominal thickness and approximate weight :-

| <u>NORMAL THICKNESS</u> | <u>APPROX. WEIGHT</u> |
|-------------------------|-----------------------|
| 3 mm                    | 7.5 kg/sq.m.          |
| 4 mm                    | 10.0 kg/sq.m.         |
| 5 mm                    | 12.5 kg/sq.m..        |
| 6 mm                    | 15.0 kg/sq.m.         |
| 8 mm                    | 20.0 kg/sq.m.         |
| 10 mm                   | 25.0 kg/sq.m.         |
| 12 mm                   | 30.0 kg/sq.m.         |
| 15 mm                   | 37.5 kg/sq.m.         |
| 19 mm                   | 47.5 kg/sq.m.         |
| 25 mm                   | 63.5 kg/sq.m.         |

J. Wired Glass

1. Wired glass shall be cast wired glass or polished wired glass with wire completely embedded and shall be either of the following :
  - a) Georgian wired cast glass - a translucent glass with a cast patterned surface, 13mm square wire mesh, electrically welded at each intersection, 6mm nominal thickness and weight approximately 17 kg/m2.
  - b) Georgian wired polished plate - a transparent glass with both faces ground and polished, 13mm square wire mesh, electrically welded at each intersection, 6mm nominal thickness and weight approximately 17 kg/m2.
  - c) One way wired glass - a proprietary glass wired in one direction only, with wires parallel and at approximately 25mm centres.

K. Heat Strengthened Glass/ Tempered Glass

1. Heat strengthened glass shall be an annealed glass that has been heat treated to approximately double the strength and impact resistance of untreated glass but not to the requirements of safety glass.
2. Heat strengthened glass shall be to ASTM C 1048 with surface compression of the glass in the range of 2400 - 45000 kPa.
3. Tempered glass shall be a fully tempered processed flat safety glass conforming to ANSI Z97-1: 1984. Tempered glass shall be a safety glass that has been heat treated to approximately four times the strength and impact resistance of untreated glass and upon fracture would break into small comparatively harmless fragments.
4. If located in areas liable to heat build-up, tempered glass shall be heat soak tested to match its intended use.
5. Sizes of all tempered glass required shall be predetermined and ordered to the correct size. Once the glass has been toughened it cannot be cut or worked. It is the contractor's responsibility to check and confirm dimensions of glass prior to ordering. No attempt shall be made to cut or work tempered glass on site.
6. The length to breadth ratio of the tempered glass shall be not greater than 7:1.
7. Provide heat strengthened glass and/or tempered glass where required by the glass product selected, design wind pressures, anticipated thermal stress. Where practical increased glass thickness is to be provided to meet the requirements of design pressures, subject to architect endorsement of appearance variation caused thereby.
8. Heat strengthened and tempered glass shall be examined by the glass manufacturer to detect and discard any lights which exceed the following tolerances: 1.5mm bow in 600mm; 3mm bow in 1500mm; 6mm bow in 3000mm; 9mm bow in 4500mm. Where the strengthening process results in essentially

parallel ripples or waves, the deviation from flatness at any peak shall not exceed 0.13mm, and the difference between adjacent peaks shall not exceed 0.08mm. Where bow tolerance and wave tolerance differ, the stricter requirements shall govern. Direction of ripples shall be consistent and in conformance with architectural design.

9. All tempered glass shall be heat soak tested or subjected to other quality control measures acceptable to the Architect to minimize the occurrence of nickel sulphide stones. This specification defines nickel sulphide stones as a glass material defect. Installed tempered glass which breaks due to nickel sulphide stones shall be considered a material defect and shall be replaced (at no charge).
10. All tempered glass proposed by the contractor shall receive "Heat Soak Test" at a temperature of 240 degree C for three (3) hours. Provide Architect with written record of heat soak test.
11. Record shall indicate test conditions, date of test, number and size of lights tested per load, spacing of lights in oven, time glass was in the heat soak furnace, test results (i.e. number of light which broke).
12. The glass proposed by the Contractor shall be of a thickness not less than the specified minimum thickness and shall have structural integrity and shall not have any harmful scratches, pinholes and uneven, sharp angled or filed edge.

#### L. Tinted Glass

1. Unless otherwise specified, all tinted glass shall be body tinted where the colouring matter extends evenly through the whole thickness of the glass. Colour formed by surface coating is not acceptable.
2. Tinted glass shall be sheet, float or plate glass for solar control glazing with the required degree of light transmission and thickness as specified in the particular specification or on drawings.
3. The colour of the tint shall be submitted and approved by the Architect.
4. Tinted glass shall be of an approved proprietary glass which is of :-
  - a) Body tinted float glass, or
  - b) Body tinted cast glass, or
  - c) Body tinted solar control tempered glass as specified.

#### M. Laminated Glass

1. Laminated glass shall be an approved proprietary glass consisting of two or more panes of glass bonded together and incorporating an interlayer of plastic reinforcing material between each pane to ensure the performance specified. The interlayers are resistant to penetration and they absorb impact shock, holding the glass in place and preventing extensive spacing of glass fragments.
2. Laminated safety glass shall comply with ANS1 Z97.1: 1984.
3. Laminated anti-bandit security glass shall comply with B.S. 5544: 1978 (1994).
4. Laminated glass can usually be cut to size and worked after manufacture and shall be carried out in strict accordance with manufacturer's instructions.
5. Bullet-resistance laminated security glass shall comply with B.S. 5051: Part 1: 1988 (1994) and to be used to provide protection against attack with firearms. The thickness and construction depends on the type of weapon and ammunition against which protection is required.

N. Tempered Laminated Glass

1. Tempered laminated glass shall conform to ANSI Z97.1 1984 and BS Code of Practice BS 6262: 1982.
2. Pattern of the ceramic-frit on the tempered laminated glass at the canopy and any other location as indicated on the drawings shall be submitted for the Architect's approval.

O. Opaque Glass

Opaque glass shall be clear float glass coated on one side with a coloured ceramic and then fixed and toughened to produce a totally opaque glass with no light transmission with thickness and colour as specified.

P. Mirror Glass:

1. All mirror glass shall be selected tempered glass suitable for silvering.
2. Mirror glass shall be 8mm thick min., unless otherwise specified
3. Edges of mirror glass shall be polished square or bevelled as specified.
4. Mirror glass shall be backed with an even coating of silver and two (2) coats of shellac varnish or other approved treatment to resist moisture and corrosion.

**2.03 ANCILLARY MATERIALS**

A. Silicone Rubber: ASTM C920, class and use as best suited for the intended purpose; Provide one of the following:

B. Glazing Sealants

1. Glazing sealant to the joint between the glass and metal bead or butt joint between glass panes shall be one-part gun grade silicone-based glazing sealants of translucent colour conforming to B.S. 5889: 1989.
2. Glazing Sealant: Sealant colours are to be custom colours selected by the Architect. All Sealants shall be test verified for use by compatibility adhesion, and other tests as stated in Section 07 92 00-Joint Sealants prior to being used.

C. Setting and Location Blocks

1. Setting and location blocks shall be dense extruded neoprene, silicone or EPDM.
2. ASTM C864; neoprene blocks, 80 to 90 Shore A durometer hardness and at least 55mm for each square meter of glass, or of sufficient length to limit imposed load to 414 KPa. Each block shall be 100mm in length minimum. Setting blocks shall be compatible with glazing sealant.
3. Neoprene and EPDM blocks are acceptable for insulating glass units only when permitted by the fabricator of these units.
4. Shims shall be of the same material, hardness, length and width as the setting blocks.

5. Dense gaskets shall comply with ASTM C 864. External gaskets shall be neoprene or Santoprene. Internal gaskets shall be neoprene, santoprene or EPDM. Reversible gaskets for re-glazing if used shall not be EPDM.
6. Sponge gaskets shall comply with ASTM C 509 and designed to provide 20-35% compression.
7. Gasket profiles shall be designed to produce a glass edge pressure of between 0.7N/mm minimum and 1.75 N/mm maximum.
8. Gaskets shall be hot melt injection moulded at corners where compatible with installation procedures.
9. Lead blocks shall be used on very large, thick plate/float glass to limit imposed load to 827KPa. Lead block length shall be 27mm for each square meter of glass. Each block shall be 100mm in length minimum. Lead blocks shall only be used for large lites of heavy float/plate glass, and not for insulating, laminated or wire glasses.
10. Silicone setting blocks of 90 Shore A durometer hardness are required where structural silicone occurs at sill. They are acceptable for insulating glass units with silicone edge seals.
11. Setting blocks and chairs shall be secured against migration.

D. Edge Blocks:

1. ASTM C864; neoprene blocks, 60 to 70 Shore A durometer hardness dense neoprene, silicone or EPDM. Install block with clearance between block and bearing surface. Positively secure blocks in position. 76mm minimum length for all side blocks. Side and centering blocks shall be of the same material as setting blocks.
2. Side blocks are not required where glass is supported along the vertical edges with structural silicone.
3. Extruded silicone side blocks are acceptable for insulating glass units with silicone edge seals.

E. Mirror Mastic: An adhesive setting compound, manufactured specifically for setting mirrors by spot application, certified as compatible with back of mirror coating by organic protective coating manufacturer, scheduled mirror substrate and approved by mirror manufacturer. Comply with mirror mastic manufacturer's written instructions for size, number and pattern of mastic spots required for installation (Refer to Section 07 92 00 – Joint Sealants).

F. Mirror Edge Sealant: A mirror edge sealant manufactured specifically for sealing mirror edges and that has proven to be compatible with mirror coating and approved by mirrored glass manufacturer for use in protecting against silver deterioration at mirror edges.

G. Stainless Steel Angle Mirror Support and Frame: AISI Type 302 or 304; hot or cold rolled angle; 25mm x 25mm x 3mm thick; polish edge of exposed portion of angle to bright finish.

H. Backer Rods:

1. Non-gassing, closed cell or jacketed foam rods of polyolefin, polyethylene, butyl, neoprene, polyurethane, or vinyl, compatible with glazing sealant. Cellular neoprene rod where recommended by FGMA Glazing Manual.



2. Nomaco "Sof Rod" : Extruded polyolefin foam made of a nonabsorbing outer skin which does not out-gas when ruptured. Sof Rod is chemically inert and resistant to oil, gasoline and most solvents. Conforms to ASTM C962, Type A joint filler.
3. Physical Properties of SOF Rod

| Property                           | Nominal Value                   | Test Method                  |
|------------------------------------|---------------------------------|------------------------------|
| Density                            | 40kg/m <sup>3</sup>             | ASTM D1622                   |
| Tensile Strength                   | 241Kpa                          | ASTM D1623                   |
| Water Absorption                   | <0.02 gms/cc                    | ASTM C24                     |
| Compression<br>(Deflection at 50%) | 20.7Kpa                         | ASTM D5249                   |
| Compression<br>(Recovery at 50%)   | 95%                             | ASTM D5249                   |
| Outgassing                         | None                            | ASTM C24                     |
| Permeability                       | Rod 1.7 Perms<br>Joint 50 Perms | ASTM E-96<br>(at 23OC, 0%RH) |
| Temperature Range                  | -43° F to 91° C                 | -----                        |

- I. Joint Cleaner, Primer and Sealer: The products recommended by the manufacturer. Never apply or allow primers to come into contact with glass surfaces.

J. Gaskets and Weather-strips

1. Gaskets and weather-strips, except at structural silicone glazing, shall conform as a minimum to BS 6262 except :-
  - a) Sponge gaskets shall be extruded black neoprene with a hardness of 40+/-5 durometer Shore A and conforming to ASTM C 509, ASTM C542, and ASTM D2000. Design sponge gaskets to provide 20% to 35% compression.
  - b) Dense gaskets shall be black extrusions with a Shore A hardness of 75 +/-5 for hollow profiles and 60 +/-5 for solid profiles, and conforming to ASTM C 864. Outdoor gaskets shall be neoprene or Santoprene. Indoor gaskets shall be neoprene, Santoprene or EPDM. Where indoor and outdoor gaskets are reversible for reglazing, EPDM shall not be used for either gasket.
  - c) Injection mold all corners of gaskets where compatible with installation procedures.
  - d) Interior and exterior gasket profiles shall be designed to produce a glass edge pressure of not less than 0.70N/mm, nor more than 1.75 N/mm.

K. Putty

1. Putty for glazing to softwood and absorbent hardwood shall be in accordance with B.S. 544 : 1969 (1994), prepared from the best washed whiting and boiled linseed oil, well kneaded together, and with a proportion of not less than 10% of white lead ground in oil worked into it during preparation.
2. Putty for glazing to metal and non-absorbent hardwood shall be an approved proprietary brand recommended by the glass supplier or window manufacturer for the particular application and with setting properties and unpainted life to suit the construction programme.
3. Non-setting compound shall be an approved brand recommended by the window manufacturer for the particular application.
4. Compounds for glazing plastic sheets shall be comparable with the proprietary sheets.

L. Wash Leather

All glazing to wood doors shall be bedded in best quality soft wash leather. The leather shall be cut into strips of the necessary widths and lengths and placed in position with glass and glazing beads so that the whole is firm but not so tight as to cause the glass to crack when doors are slammed normally.

M. Preformed Compression Gaskets

Preformed compression gaskets shall be neoprene gaskets which are formulated to withstand stretching, compression, ageing and to minimise stiffening at low temperature.

## **2.04 FABRICATION**

- A. Sizes: Fabricate glass to sizes required for glazing openings indicated, with edge clearances and tolerances complying with recommendations of glass manufacturer. Provide thicknesses indicated or, if not otherwise indicated, as recommended by glass manufacturer for application indicated.
- B. Cutting: Obtain sizes from shop drawings or by field measurement. Cut glass to fit each opening with edge clearances and bite on glass as recommended by glass manufacturer. Do not nip glass edges. Factory cut heavy heat absorbing glass (9mm and above) and heavy float glass (13mm and above). Edges may be wheel cut or sawed and seamed at manufacturer's option. For glass to be cut at site, provide glass larger than required so as to obtain, clean-cut edges without seaming or nipping. Do not cut, seam, nip or abrade glass after heat-tempering.
- C. Edgework for Butt Glazing: Where glass is to be butt joined with silicone sealant, provide flat ground butting glass edges having a satin finished flat edge with eased arrise corners.
- D. Edgework for Exposed Edges: Where edges of glass are to be exposed in the finish work, provide ground and polished edges and slightly eased arrises and eased corners.

## **PART 3 EXECUTION**

### **3.01 EXAMINATION / INSPECTION**

Examine substrates, adjoining construction and conditions under which the Work is to be installed. Do not proceed with the Work until unsatisfactory conditions have been corrected.

1. Prior to any glazing work, carefully examine glass framing members. The glazing framing members must be square, true, and properly designed to resist any load they will have to support. Glass surround shall not impose any bending or high concentrated compressive loads on the glass. All framing members shall not deflect more than 1/175 of its span, with a maximum of 10mm at any point.
2. The twisting of the horizontal bottom member due to the dead weight of the glass shall be limited to 0.50 from the horizontal plane. Glazing stops, or any other fitting, shall be designed to resist any load transmitted by the glazing.
3. Framing members that have off-sets at the corners, or are excessively racked, twisted or bowed shall be rejected and replaced.
4. Defective or damaged glass framing members shall be replaced with new at no additional cost to the Employer.

### **3.02 PREPARATION**

- A. Clean glazing channels, stops and rabbets to receive glazing materials of obstructions and deleterious substances which might impair the work. Remove protective coatings which might interfere with bond of sealants. Comply with manufacturer's instructions for final wiping of surfaces immediately before application of primer and glazing compounds or tapes.
- B. Prime surfaces to receive glazing compounds, except where manufacturer provides written reports of tests conducted by an independent testing agency which demonstrates that primer or sealer is not required for the conditions of use and the substrates involved. When priming, comply with manufacturer's recommendations.
- C. Provide powered scaffold, hose, water supply, powered fans, manpower and any other equipment, etc. to perform all glass installation and field testing of glazing and windows.
- D. Under no circumstances shall defective or ex- expired glass, sealants, and glazing materials be used.

### **3.03 INSTALLATION**

- A. Standards: Unless otherwise shown or specified, comply with recommendations and requirements of the GANA "Glazing Sealing Systems Manual" and "Glazing Manual". For the installation of tinted or spandrel glass comply with glass manufacturer's recommendations.
- B. Inspect each piece of glass immediately before installation. Do not install any pieces which are improperly sized or have damaged edges, scratches or abrasion or other evidence of damage. Remove labels from glass immediately after installation.
- C. Locate setting blocks at sill one-quarter of the width in from each end of the glass, unless otherwise recommended by the glass manufacturer. Use blocks of sized 3mm wider than the glass thickness and 1.6mm to 3mm less than the width of the glazing channel to support the glass and in accordance with manufacturer's recommendations.

- D. Provide face shims for all glass sizes larger than 50 united in., to separate glass from stops, except where continuous glazing gaskets or felts are provided. Locate face shims opposite each other and no farther than 60cm apart and no closer than 30cm to a corner. Make bite of spacer on glass a nominal 6mm or greater.
- E. Provide edge blocks, located in glazing rebate as recommended by the glass manufacturer, to insure against displacement of the glass and against metal to glass contact within the rebate and to ensure permanently adequate bite of the glass within the glazing system.
- F. Set glass in a manner which produces greatest possible degree of uniformity in appearance. Where safety glazing is scheduled or required, install glass after detaching removable safety glazing label unless otherwise required by authorities having jurisdiction. If local authorities require permanent labeling, install glass with permanent safety glazing label in concealed or inconspicuous locations subject to selection by the Architect.
- G. Do not use 2 different glazing materials in the same joint system unless the manufacturer of each material has stated in writing that his material is fully compatible with the other material.
- H. Use suitable protection to limit coverage of glazing materials to the surfaces intended for sealants.
- I. Tool exposed surfaces of glazing materials to provide a slight wash away from the glass. Install exposed tapes and gaskets with a slight protrusion above stops in the final compressed condition.
- J. Except as otherwise specified, comply with BS6262. Provide a minimum nominal glass bite of 13 mm. Where joint movement will result in variable glass bite, increase nominal bite to provide 10.0mm minimum bit and 6.0 mm minimum edge clearance. In no case shall the front and back clearances be less than 5.0 mm.
- K. Mirror Glazing
  - 1. Adhesive Mounting of Mirrors
    - a. Identify and examine surfaces over which mirror is to be mounted. Comply with manufacturer's printed installation directions for preparation of mounting surfaces including coating surfaces with mastic manufacturer's special bond coating where applicable or priming and sealing substrate as recommended.
    - b. Paint back of mirror with an additional coat of moisture-resistant paint of the type recommended by the mirror manufacturer and compatible with mirror mastic and substrate. Apply mirror edge sealer to all edges of mirror in accordance with manufacturers' instructions and recommendations.
    - c. Provide 13mm thick fire-retardant treated plywood mirror backerboards. Provide 25mm x 25mm x 3mm. thick stainless steel angle frame with mitered corners and bright polished finished exposed edges. Secure angle frame to wall construction with fastening devices appropriate for substrates encountered spaced 40cm. o.c. maximum. Secure plywood backerboard to wall using fastening devices appropriate for substrates encountered spaced 300mm maximum at perimeter 13mm from corners and three rows of 3 fasteners each in the backerboard field. Countersink fasteners flush with plywood surface. Butt adjacent panels without lapping. Prepare panels for priming as required by manufacturer of mirror mastic.

- d. Support mirror with stainless steel satin finished angle frame with bottom support designed to withstand mirror weight. Provide 3mm thick by 10cm long x 6mm wide setting blocks at quarter points. Apply mirror mastic in accordance with mastic manufacturer's instructions utilizing mastic manufacturer's special tool to assure complete and accurate coverage. Do not cover more than 25% of mirror back unless otherwise recommended by the mastic manufacture. Provide neoprene shims, double face tape or other type compatible material to allow for minimum clearances for mastic. Apply mirror to substrate so that areas not covered with mastic will remain open for ventilation, with minimum clearance from substrate as recommended by mastic manufacturer. Provide temporary rigid support until adhesive has set.
  - e. Fill 3mm wide joint between stainless steel angle and face of mirror with continuous backer rod and clear silicone sealant.
- 2. Channel Glazing of Mirrors: Use interior glazing system specified herein and comply with mirror manufacturer's instructions and recommendations.
- L. Sealant:
  - 1. Refer to Section 07 92 00: Joint Sealants for other sealant and application requirements.
  - 2. Provide glazing sealants and gaskets as required for particular glazing application. Coordinate with other Sections for material compatibility.
  - 3. Provide filler rod where sealants are used in the following locations:
    - (a) Head and jamb channels.
    - (b) Clear glass over 800 sq. cm in size.
  - 4. Protect all adjoining surfaces to sealants and gaskets against staining. Clean, prime and mask at structural silicone joints within 8 hours before applying the silicone sealant.
  - 5. Joints, joint surfaces and glazing rebates shall be clean, dry, and free of any material that may have an adverse effect on the bonding or sealing of the sealant and gasket materials, or on the proper drainage of the glazing rebates.
  - 6. Mask all joints prior to installation of sealant. Apply sealants and gaskets under the conditions, and in the manner, recommended by the manufacturer. No sealant that has started to set in its container or has exceeded its shelf/pot life shall be used.
  - 7. The sealant manufacturer shall provide written notification if the relevant substrates in contact with sealants will, or will, not require priming. The recommendations will be based on tests conducted by the sealant manufacturer using samples of the relevant substrates for the Project. Unless printed instructions state to the contrary, sealant shall not be applied when substrates are wet
  - 8. Fill all joints continuously and completely with sealant, forming a neat, uniform, concave or flat bed finish flush with adjoining surfaces, unless otherwise shown on the drawings. All sealant surfaces shall be tooled smooth, using dry tool technique, Use of water, soap water or tooling lubricant is strictly prohibited.
  - 9. Exposed sealants shall be installed so that top surfaces of the horizontal sealant beads are sloped to drain water away.

10. Do not leave voids in glazing channels except as specifically indicated or recommended by glass manufacturer. Force sealant into channel to eliminate voids. Tool exposed surfaces to slight wash away from joint. Trim and clean promptly.
11. Do not allow sealant to close weeps of aluminum storefront or window system.
12. No fasteners shall penetrate glazing pockets.
13. Install protective glass edge trims at all exposed glass edges fixed with silicone sealant.

M. Gaskets:

1. Provide adequate anchorage, particularly for driven-in wedge gaskets.
2. Mitre and weld ends of channel gaskets at corners to provide continuous gaskets.
3. Seal face gaskets at corners with sealant to close opening and prevent withdrawal of gaskets from corners. Corner gaskets shall be pre-moulded in L-shape configuration.
4. Glazing gaskets shall be of the profile, dimension and durometer hardness required to support the glass and resist water penetrations. Exterior glazing gaskets (if used) shall have factory made pre-molded corners.
5. Incorrect placement, levelling or lining-through of angles to insitu finishings, and the like fixed by others, shall be immediately reported to the Employer's Project Manager.
6. Coordinate and carry out the installation of the glazing in sequence with related work operations to be executed by others as necessary to achieve a weather-tight and satisfactory visual relationship between the various components and the structure, finishes and weather seals.
7. Compatibility, adhesion and staining tests shall have been proved positive before any sealants may be installed on the Work. Coordinate with other trades to assure compatibility of intersecting sealants.
8. Temporarily clamp glass during cure of structural silicone. After sufficient cure, remove clamps and fill any gaps in silicone as recommended by sealant manufacturer.
9. Mask glass, aluminium, and other substrate, during application of structural silicone sealant. Remove masking immediately after tooling sealant.
10. Structural silicone shall not be applied to edges of insulating glass units, or to edges of laminated glass units. Sealants used as weather seals shall be adhered to, or placed against, the edge of a laminated glass unit inter-layer.
11. Before commencing sealant installation, check to Ensure that joint dimensions are within limits specified for the sealant, the surfaces are smooth and undamaged, and that the preparatory work, which must be done before assembly or sealing of the joint has been carried out.
12. Inform the Architect if joints are not suitable to receive sealant and submit proposals for rectification.

N. Glass To Glass Glazing:

Apply structural type silicone rubber sealant to abutting surfaces of glass in accordance with manufacturer's recommendations.

### **3.04      CLEANING**

- A. Clean excess sealant or compound from glass and framing members immediately after application using solvents or cleaners recommended by manufacturers.
- B. Wash glass on both faces not more than 4 days prior to acceptance by the Owner. Comply with instructions and recommendations of the glass manufacturer and glazing materials manufacturer for cleaning in each case.
- C. Periodically remove from the site debris, excess materials and unused tools and equipment resulting from this work. At the conclusion of construction, leave the premises in a clean condition.

### **3.05      PROTECTION**

- A. Protect glass from breakage immediately upon installation. Use streamers or ribbons suitably attached to framing and held free of the glass. Do not apply warning markings directly to the glass. Do not permit edges of glass to be exposed to standing water.
- B. Protect glass and glazing materials during the construction period so that they will be without any indication of damage or deterioration at the time of acceptance by the Employer. Cover glass to protect it from activities that might abrade the surfaces.
- C. Replace glass during the construction period which is broken, cracked, chipped or damaged in any way and from any source, including weather, vandalism or accidents.
- D. Maintain glass in a reasonably clean condition during construction so that it will not become stained and will not contribute to the deterioration of glazing materials. Conduct a systematic inspection program not less than once a month for glass. Clean to meet above requirements.

**END OF SECTION**